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Development of a Personalized Lip color Recommendation System: A Review and Color Similarity-Based Approach

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Background

When purchasing lip products, people often discover specific shades that suit them well or become their favorites. However, with new products being launched every season and an overwhelming variety of colors available, finding a shade similar to a beloved product is not always easy.
Moreover, since selecting a color is a highly subjective process, many people experience disappointment after purchasing a seemingly similar shade that does not meet their expectations. Such purchasing failures are a common experience.

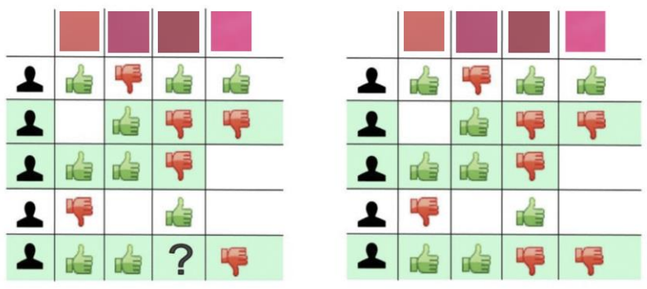
In addition, favorite products can be unexpectedly discontinued, or users may want to try a different formula in the same shade, which presents another challenge. To address these issues, we conceptualized and developed this Project.

The Project is a system where users input their favorite lip products and answer a few simple questions about their preferences. Based on this information, the system recommends five lip products with similar shades.
This is achieved by analyzing lip product reviews and comparing color similarities. Through this system, users can easily find alternatives that closely match their preferred shades while also making choices aligned with their personal preferences, resulting in a more satisfying experience.

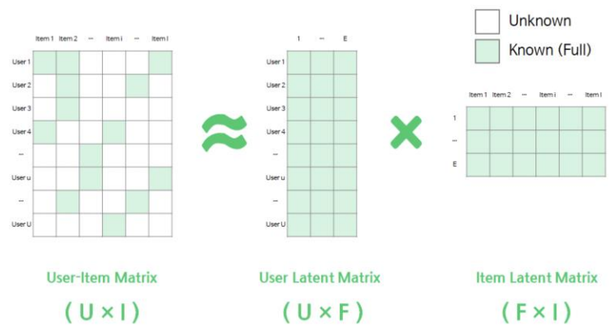
The Project goes beyond merely recommending shades; it carefully considers users' needs and tastes to assist in selecting new products. This makes it especially beneficial for users seeking alternatives to discontinued products or wanting to explore the same shade in a different formula.

Methods

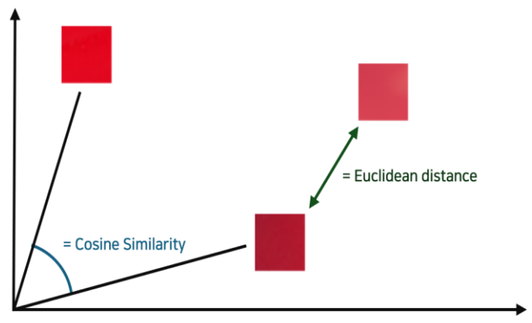
· Collaborated filtering



· Matrix Factorization, SGD(Stochastic Gradient Descent)



· Euclidiean distance, Cosine Similarity



→ **Collaborative Filtering** was utilized to predict ratings based on the similarity between users and items in the review data. Using a **Matrix Factorization** technique, specifically **Stochastic Gradient Descent (SGD),** the user-item relationships were decomposed into latent factors to achieve more precise rating predictions.
For the top-ranked products, **Euclidean Distance** and **Cosine Similarity** were applied to analyze the similarity of product attributes based on color. This quantitative evaluation was used to identify products with high color similarity.

The system was designed to recommend products tailored to user preferences and conditions, providing personalized satisfaction while balancing the diversity and similarity of product attributes.

Results

Input
1. Product color that satisfies your needs
2. Desired Formulation
3. Presence or absence of allergy (Lithol Rubine BCA)

Output
The system generates a list of recommended products,
Detailing their color, Price, Formulation, and Personal color types.

0.992801 0.985882 0.983058 0.981164 0.980134

· You can determine your personal color based on your existing products and choose colors that suit you.
· You can get more accurate results compared to subjective visual assessments.

Conclusions

01 Data Collection
The data collection method was conducted manually and was time-consuming, but automation can improve efficiency and accuracy.

02 UI Design
The current UI is simple due to technical limitations, but a more intuitive and user-friendly design can enhance accessibility and usability.

03 Category Expansion
While the system focuses on lip products, expanding to other cosmetic categories can offer more comprehensive recommendations.

04 Personal Color Diagnosis
This system helps individuals easily identify suitable colors, reducing the time and cost of professional analysis.

05 User Feedback
Adding actual user reviews of recommended products can improve reliability, making recommendations more meaningful and valuable.

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